

Erdős Problem #883

EP-883

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

For $A \subseteq \{1, \dots, n\}$ let $G(A)$ be the graph with vertex set A , where two integers are joined by an edge if they are coprime.

Is it true that if $|A| > \lfloor \frac{n}{2} \rfloor + \lfloor \frac{n}{3} \rfloor - \lfloor \frac{n}{6} \rfloor$ then $G(A)$ contains all odd cycles of length $\leq \frac{n}{3} + 1$?

Is it true that, for every $\ell \geq 1$, if n is sufficiently large and $|A| > \lfloor \frac{n}{2} \rfloor + \lfloor \frac{n}{3} \rfloor - \lfloor \frac{n}{6} \rfloor$ then $G(A)$ must contain a complete $(1, \ell, \ell)$ tripartite graph on $2\ell + 1$ vertices?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>